

PRAKTIKUM BASIS DATA

MODUL 13

NoSQL Document Database

JURNAL



Oleh:

Rafael Putra Septava

103122400015

PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

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JURNAL

1) Tambahkan 1 toko baru bebas ke koleksi toko, dengan ketentuan:

- Minimal memiliki 2 produk di dalam array products
- Isi semua field: name, category, city, address, products

```
jurnal13> db.toko.insertOne({
|   "name": "Toko Arch",
|   "category": "Elektronik",
|   "city": "Yogyakarta",
|   "address": {
|       "street": "Jl. Malioboro",
|       "zipcode": "55213"
|   },
|   "products": [
|       {
|           "item": "Arch Speaker Bluetooth",
|           "price": 750000,
|           "stock": 15
|       },
|       {
|           "item": "Arch Smart TV",
|           "price": 6500000,
|           "stock": 6
|       }
|   ]
| })
{
  acknowledged: true,
  insertedId: ObjectId('6a0e9cd43b2c94b1799df8a7')
}
```

- 2) Tampilkan semua toko yang berada di kota Jakarta.

```
jurnal13> db.toko.find({ city: "Jakarta" })
[
  {
    _id: ObjectId('6a0e9ca73b2c94b1799df8a3'),
    name: 'Toko Berkah',
    category: 'Elektronik',
    city: 'Jakarta',
    address: { street: 'Jl. Sudirman', zipcode: '10220' },
    products: [
      { item: 'TV', price: 3500000, stock: 10 },
      { item: 'Kulkas', price: 2800000, stock: 5 }
    ]
  },
  {
    _id: ObjectId('6a0e9ca73b2c94b1799df8a5'),
    name: 'Toko Salim Bakery',
    category: 'Elektronik',
    city: 'Jakarta',
    address: { street: 'Jl. Sudirman', zipcode: '10220' },
    products: [
      { item: 'Monitor LED', price: 7700000, stock: 11 },
      { item: 'Airfryer Panasonic', price: 4850000, stock: 7 }
    ]
  }
]
```

- 3) Tampilkan toko yang memiliki produk dengan price lebih dari 3.000.000.

Hint:

- Gunakan dot notation: products.price • Gunakan operator: \$gt

```
jurnal13> db.toko.find({
|   "products.price": { $gt: 3000000 }
| })
[
  {
    _id: ObjectId('6a0e9ca73b2c94b1799df8a3'),
    name: 'Toko Berkah',
    category: 'Elektronik',
    city: 'Jakarta',
    address: { street: 'Jl. Sudirman', zipcode: '10220' },
    products: [
      { item: 'TV', price: 3500000, stock: 10 },
      { item: 'Kulkas', price: 2800000, stock: 5 }
    ]
  },
  {
    _id: ObjectId('6a0e9ca73b2c94b1799df8a4'),
    name: 'Toko Maju',
    category: 'Elektronik',
    city: 'Bandung',
    address: { street: 'Jl. Asia Afrika', zipcode: '40111' },
    products: [
      { item: 'HP', price: 1500000, stock: 20 },
      { item: 'Laptop', price: 8000000, stock: 3 }
    ]
  },
  {
    _id: ObjectId('6a0e9ca73b2c94b1799df8a5'),
    name: 'Toko Salim Bakery',
    category: 'Elektronik',
    city: 'Jakarta',
    address: { street: 'Jl. Sudirman', zipcode: '10220' },
    products: [
      { item: 'Monitor LED', price: 7700000, stock: 11 },
      { item: 'Airfryer Panasonic', price: 4850000, stock: 7 }
    ]
  },
]
```

```

{
  _id: ObjectId('6a0e9ca73b2c94b1799df8a6'),
  name: 'Toko Sudinotech',
  category: 'Elektronik',
  city: 'Surabaya',
  address: { street: 'Jl. Sudirman', zipcode: '10220' },
  products: [
    { item: 'RTX 55000', price: 8500000, stock: 20 },
    { item: 'Set PC Gaming Radeon', price: 12000000, stock: 9 }
  ]
},
{
  _id: ObjectId('6a0e9cd43b2c94b1799df8a7'),
  name: 'Toko Arch',
  category: 'Elektronik',
  city: 'Yogyakarta',
  address: { street: 'Jl. Malioboro', zipcode: '55213' },
  products: [
    { item: 'Arch Speaker Bluetooth', price: 750000, stock: 15 },
    { item: 'Arch Smart TV', price: 6500000, stock: 6 }
  ]
}
]

```

- 4) Ubah category Toko Berkah dari Fashion menjadi Aksesoris. Tambahkan 1 produk baru ke dalam array products milik Toko Maju menggunakan \$push.

Hint:

- Filter berdasarkan name: Toko Maju
- Gunakan operator \$push untuk menambah elemen ke array

```
jurnal13> db.toko.updateOne(
|   { name: "Toko Berkah" },
|   {
|     $set: {
|       category: "Aksesoris"
|     }
|   }
| )
|
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}

jurnal13> db.toko.updateOne(
|   { name: "Toko Maju" },
|   {
|     $push: {
|       products: {
|         item: "Mouse Gaming",
|         price: 450000,
|         stock: 12
|       }
|     }
|   }
| )
|
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

- 5) Hitung jumlah toko per city, urutkan dari kota yang paling banyak tokonya. Gunakan:

- \$group dengan \$sum: 1
- \$sort descending (-1)

```
jurnal13> db.toko.aggregate([
|   {
|     $group: {
|       _id: "$city",
|       jumlah_toko: { $sum: 1 }
|     }
|   },
|   {
|     $sort: {
|       jumlah_toko: -1
|     }
|   }
| ])
[
|   { _id: 'Jakarta', jumlah_toko: 2 },
|   { _id: 'Surabaya', jumlah_toko: 1 },
|   { _id: 'Bandung', jumlah_toko: 1 },
|   { _id: 'Yogyakarta', jumlah_toko: 1 }
| ]
```

- 6) Hitung rata-rata harga produk per category. Tahapan pipeline:

- \$unwind: '\$products' (bongkar array products dulu)
- \$group: kelompokkan per category, hitung \$avg dari products.price
- \$sort: urutkan sesuai kebutuhan

```
jurnal13> db.toko.aggregate([
|   {
|     $unwind: "$products"
|   },
|   {
|     $group: {
|       _id: "$category",
|       rata_rata_harga: {
|         $avg: "$products.price"
|       }
|     }
|   },
|   {
|     $sort: {
|       rata_rata_harga: -1
|     }
|   }
| ])
|
| [
|   { _id: 'Elektronik', rata_rata_harga: 5583333.333333333 },
|   { _id: 'Aksesoris', rata_rata_harga: 3150000 }
| ]
```

- 7) Tambahkan 1 produk baru ke dalam array products milik Toko Maju menggunakan \$push.

Hint:

- Filter berdasarkan name: Toko Maju
- Gunakan operator \$push untuk menambah elemen ke array


```

jurnal13> db.toko.updateOne(
|   { name: "Toko Maju" },
|   {
|     $push: {
|       products: {
|         item: "Keyboard Mechanic",
|         price: 950000,
|         stock: 8
|       }
|     }
|   }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}

```

8) Tampilkan harga produk tertinggi (\$max) per city.

Hint:

- Unwind products terlebih dahulu
- Gunakan \$max pada products.price

```
jurnal13> db.toko.aggregate([
|   {
|     $unwind: "$products"
|   },
|   {
|     $group: {
|       _id: "$city",
|       harga_tertinggi: {
|         $max: "$products.price"
|       }
|     }
|   }
| ])
|
| [
|   { _id: 'Surabaya', harga_tertinggi: 12000000 },
|   { _id: 'Yogyakarta', harga_tertinggi: 6500000 },
|   { _id: 'Jakarta', harga_tertinggi: 7700000 },
|   { _id: 'Bandung', harga_tertinggi: 8000000 }
| ]
```

9) Buat pipeline lengkap dengan 4 tahap:

- \$match — filter hanya toko category Elektronik
- \$unwind — bongkar array products
- \$group — hitung total stock per city
- \$sort — urutkan dari total stock terbesar

```

jurnal13> db.toko.aggregate([
|   {
|     $match: {
|       category: "Elektronik"
|     }
|   },
|   {
|     $unwind: "$products"
|   },
|   {
|     $group: {
|       _id: "$city",
|       total_stock: {
|         $sum: "$products.stock"
|       }
|     }
|   },
|   {
|     $sort: {
|       total_stock: -1
|     }
|   }
| ])
|
| [
|   { _id: 'Bandung', total_stock: 43 },
|   { _id: 'Surabaya', total_stock: 29 },
|   { _id: 'Yogyakarta', total_stock: 21 },
|   { _id: 'Jakarta', total_stock: 18 }
| ]

```

10) Hapus toko yang memenuhi dua kondisi sekaligus:

- category adalah Fashion
- city adalah Surabaya

Hint:

- Gunakan \$and atau tulis kedua kondisi dalam satu objek filter

```
jurnal13> db.toko.deleteOne({
|   $and: [
|     { category: "Fashion" },
|     { city: "Surabaya" }
|   ]
| })
{ acknowledged: true, deletedCount: 0 }
```